In defined contribution plans, by contrast, the employer contributes a fixed amount. Hence, the contribution is "defined." The retirement account works like a glorified bank balance and is invested in a variety of asset classes like stocks, bonds, or real estate. Investment choices are made by the employee, but the employer usually chooses the types of funds available.[20] Since returns fluctuate over time, and individuals bear the investment risk, and since some employees put in more of their own money than do others, the amount available at retirement is not fixed.

The biggest development in pensions over the last thirty years is that defined benefit plans are increasingly scarce. "Defined benefit plans are going the way of the dodo," says Olivia Mitchell, a pension and retirement expert at the Wharton School.[21] They are still the mainstay type of plan for government workers, but many corporations have frozen their old defined benefit plans and do not permit new entrants. Companies have also converted defined benefit plans to defined contribution ones.[22]

There are huge sums sitting in pension funds. Total pension assets represent over 70% of GDP in the United States. Pensions are even more important in other countries, including Australia and the Netherlands, where they represent over 90% and 135% of GDP respectively.[23]

Figure 1.2, Panel A graphs the total value of private pension fund AUM in the United States (all data in Figure 1.2 are from the Flow of Funds). At the end of 2012, there were \$6.6 trillion of private pension fund assets. Pension assets have been growing at 8.2% a year, but Panel A shows strong dips during the 2000s due to the dot-com bust and the 2008 financial crisis, when pension AUM shrank by 29%. The large declines in AUM occurred because pension funds have enthusiastically embraced risk and their increased holdings of equities and other risky assets did poorly during these periods.

Panel B of Figure 1.2 shows that the proportion of private pension fund money in non-fixed income assets rose from less than 20% in the early 1950s to exceed 70% in 2012. Most of this diversification was into U.S. equities in the 1960s, with international and emerging market equities added in the 1970s and 1980s. Since the mid-1990s, and especially over the 2000s, pension funds have scrambled to add *alternative assets*—notably hedge funds and private equity, although chapters 17 and 18, respectively, will show that these are not asset classes and

[20] This limited universe is often not in the best interest of the employee, see chapter 3.

[21] Quoted in "Defined Contribution Plans: Over to You," *Economist*, April 9, 2011.

[22] Defined benefit plans technically include *cash balance* plans, which are plans where the employers prefund contributions. From the point of view of the employee, cash balance plans look like defined contribution plans—they receive statements with their own balances, which they can withdraw when they leave the firm or retire—even though these balances are notional; the funds are pooled and managed centrally. Munnell and Soto (2007) and Rauh, Stefanescu, and Zeldes (2012) study defined benefit pension freezes and conversions to defined contribution plans.

[23] See OECD, Pension Markets in Focus, September 2012, Issue 9.